Lecture 6: LLMs for Credit Risk Analysis

Large Language Models in Finance

Juan F. Imbet

Paris Dauphine – PSL University

Outline

Why Credit AI Is Different

The Apple Card Incident (2019)

David Heinemeier Hansson's Apple Card credit limit was 20$\times$ lower than his wife's — despite joint tax returns and comparable credit profiles. Senator Warren wrote to Goldman Sachs. The NY DFS opened a formal investigation.

Credit data is simultaneously:

- The richest behavioural dataset available to financial institutions
- One of the most legally regulated datasets in existence

Getting it wrong in either direction carries consequences:

- Denying credit unfairly $\rightarrow$ Fair Lending violations
- Extending it recklessly $\rightarrow$ capital losses and systemic risk
- Both $\rightarrow$ regulatory action, litigation, reputational damage

THE BIGGER PICTURE

The core tension. Credit is the one domain where predictive accu-

Credit Bureau Data

The three major US credit bureaus (Equifax, Experian, TransUnion) each maintain files on ≈220 million adults. A standard credit file contains:

- **Account data**: 24–84 months of payment history per account, balance, limit
- **Inquiry data**: hard inquiries (applications), soft inquiries (pre-qualification)
- **Public records**: bankruptcies (7–10 years), civil judgments
- **Collections**: charged-off accounts

FICO score factor weights:

Factor	Weight
Payment history	35%
Amounts owed	30%
Length of credit history	15%
New credit	10%
Credit mix	10%

Alternative Data and Regulatory Constraints

Alternative data extends credit to thin-file consumers:

- **Bank transaction data**: cash flow signals; consent governed by Dodd-Frank §1033
- **Rent and utility payments**: largest recurring obligation yet historically invisible
- **Unstructured text**: loan purpose statements; "medical emergency" $\neq$ "consolidating payday loans"
- **Social media/behavioural data**: largely prohibited by regulators (protected-class proxy risk)

Three key regulatory frameworks:

Law	Key constraint for LLM credit models
FCRA	Adverse action must cite specific reasons (reason codes). A transformer producing a single scalar output is not inherently compliant.
ECOA / Reg B	Prohibits disparate impact on protected

Preprocessing: Three Universal Challenges

1. Anonymisation and De-identification

Quasi-identifiers create re-identification risk. Sweeney (2002): 87% of the US population is uniquely identifiable by ZIP code + date of birth + sex. Standard techniques: tokenisation, generalisation (age $\rightarrow$ age bracket), suppression, differential privacy.

2. Class Imbalance

Default rates: 1–10% over 12-month horizon. A 3% default dataset with 1M rows has 30K positives and 970K negatives. A naïve "always predict non-default" model achieves 97% accuracy.

Balanced class weight:

$$w_+ = \frac{n}{2\,n_+}, \qquad w_- = \frac{n}{2\,n_-}$$

Each class gets equal aggregate influence on the gradient regardless of the class ratio.

3. Missing Data

Three mechanisms: MCAR (mean imputation unbiased), MAR (multiple

The Credit Risk Modelling Arc

Era	Method	Key insight
1968	Altman Z-score	Linear discriminant of 5 financial ratios
1974	Merton model	Equity as call option on firm assets
1990s	Reduced-form	Default as Poisson intensity process
2015–22	Gradient boosting	Non-linear tabular features; missing values
Now	Language models	Text-native signals; thin-file borrowers

Altman Z-score:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

$Z < 1.81$: distressed; $Z > 2.99$: healthy. Estimated on *sixty-six* firms in 1968. Still used as a benchmark today.

LLMs add value in three specific scenarios:

1. Text-native features (loan purpose, credit officer notes)
2. Thin-file borrowers (alternative text data resists tabularisation)

Fine-Tuning for Default Prediction

Task formulation options:

- **Sequence classification**: serialise profile as text; BERT encoder with `[CLS]` head and sigmoid. Most widely used in published LLM credit scoring papers.
- **Causal LM with label generation**: prompt decoder-only model; extract label token probability. Scales to few-shot settings.
- **Regression on probability**: output a scalar probability directly.

Input serialisation (consistency is critical):

```
Age 34. Employment: full-time, 6 years. Income: $58K.
Bureau score: 672. Delinquencies 24m: 1. Loan purpose: debt
consolidation. Assess default risk:
```

UNDER THE HOOD

Parameter-efficient fine-tuning (LoRA). Constrain the weight update to a low-rank factorisation:

$$\Delta W = BA, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times d}, \quad r \ll d$$

Structured Generation: Implied Default Probabilities

Problem

A decoder-only LLM produces a distribution over its entire vocabulary at each position — not a single scalar probability of default. How do we extract a well-defined PD?

UNDER THE HOOD

Constrained decoding (`outlines` library, Willard et al. 2023):

1. Define an output grammar: JSON object with one key `"pd"` mapped to a float in $[0, 1]$
2. At each decoding step, intersect the model's next-token distribution with the grammar's automaton
3. Renormalise over valid tokens only:

$$P_{\text{constrained}}(x_{t+1} = v \mid x_{1:t}) = \frac{\exp(\ell_v) \cdot \mathbf{1}[v \in \mathcal{V}_{\text{valid}}]}{\sum_{u \in \mathcal{V}_{\text{valid}}} \exp(\ell_u)}$$

The model generates `{"pd":` … then produces the float *digit by digit*

Probability Calibration

Definition

A model is *perfectly calibrated* if:

$$\mathbb{E}[Y \mid \hat{P} = p] = p \quad \text{for all } p \in [0, 1]$$

Among all applications scored $\hat{p}$, the fraction that actually default is $\hat{p}$.

Calibration is assessed via **reliability diagrams**: bin the interval $[0, 1]$; plot mean predicted probability vs. observed default rate in each bin. Perfect calibration = 45-degree line.

Platt scaling: fit a logistic regression on raw model scores on a held-out calibration set:

$$\hat{p}_{\text{Platt}}(x) = \sigma(a \cdot f(x) + b)$$

$a < 1$: overconfident scores; $a > 1$: underconfident.

Isotonic regression: non-parametric; finds non-decreasing step function minimising:

n

Bounded Rationality and Household Finance

CFPB (2018)

One-third of US mortgage borrowers did not shop before accepting the first offer. The average non-shopper paid ≈$300/year more in interest. Over 30 years: **tens of thousands of dollars**.

Four sources of departure from the neoclassical benchmark:

1. **Computational complexity**: the full household optimisation is a high-dimensional stochastic control problem. Households use approximations.
2. **Information asymmetry**: specialised knowledge of credit terms, tax implications, and insurance structures is required — and mostly absent.
3. **Dual-process cognition**: high-stakes financial decisions are often made in "System 1" mode (fast, affective, heuristic-driven).
4. **Preference inconsistency**: hyperbolic discounting, loss aversion, and framing effects.

LLMs as Decision-Support Agents

Three required functions:

1. **Information extraction**: parse complex financial documents (term sheets, insurance exclusions) into structured, comparable summaries
2. **Consequence tracing**: compute multi-period financial implications of each choice given the household's stated circumstances
3. **Trade-off articulation**: present options in a format that makes relevant trade-offs legible *without nudging* toward a particular choice

Practical architecture:

Component	Role
Document ingestion layer	RAG pipeline on term sheets
Household profile module	Income, wealth, risk preferences
Consequence engine	Scenario computations
Dialogue manager	Conversational trade-off presentation

Design principle

Persona Agents: Silicon Sampling

Argyle et al. (2023)

An LLM conditioned on a detailed demographic backstory reproduces the survey response distributions of the corresponding demographic group with striking fidelity. This is an empirical observation about distributional encoding, not a claim about sentience.

Persona specification $\Pi = (D, S, \Psi, L)$:

- D: demographics (age, sex, household composition, education, region)
- S: socioeconomics (employment, income, wealth, existing debt)
- Ψ: psychology (γ risk aversion, δ time preference, financial anxiety)
- L: financial literacy (Lusardi–Mitchell 3-question battery)

Two contrasting personas:

- **Maria**: age 28, rideshare driver, \$2.4K/month variable income, \$800 savings, bureau score 584, no post-secondary education, high security preference

Persona Simulation in Action

Scenario: \$20,000 personal loan, 5 years.

- Option A: fixed 12% APR, \$444/month
- Option B: variable starting 8% APR, \$406/month, capped at 18% APR

Maria's reasoning:

> *"Option B starts lower but says adjustable. If it goes to 18%... My rent is \$1,200 and some months I barely clear \$1,800. I can't risk my payment jumping when work is slow. I'd take Option A. At least I know exactly what I owe every month."*

Thomas's reasoning:

> *"At 8%, five-year total interest ≈ €4,300; at 12% ≈ €6,600. If rates stay below ≈10% on average, Option B is cheaper. My wife's income covers fixed obligations. I'd choose Option B and set aside the monthly difference into the Livret A."*

Observation

Both responses are coherent and defensible given the persona. Neither is

Credit-Specific Performance Metrics

UNDER THE HOOD

AUROC: probability that a randomly drawn defaulter scores higher than a randomly drawn non-defaulter:

$$\text{AUROC} = P(S_+ > S_-)$$

KS Statistic: maximum separation between CDFs of defaulter and non-defaulter scores:

$$\text{KS} = \sup_s |F_+(s) - F_-(s)|$$

In practice: sort all scores, compute cumulative defaulter and non-defaulter fractions, read off maximum vertical gap. **KS > 0.40** is good; **KS > 0.60** is excellent.

Gini Coefficient (accuracy ratio):

$$\text{Gini} = 2 \cdot \text{AUROC} - 1$$

This identity is *exact*, not an approximation. Derivable from the relationship between the CAP curve and the ROC curve. Range: 0 (random

SR 11-7: Model Risk Management for LLM Credit Models

SR 11-7 (Fed / OCC, 2011): every model must be conceptually sound, monitored, independently validated, and documented. LLMs are unambiguously models under this definition.

Three pillars adapted for LLMs:

1. **Model Development**
   - Document base model provenance: training data, licence, cutoff date
   - Fine-tuning data quality: labels independently verified? Vintage relative to current credit cycle?
   - Emergent behaviour: unexpected reasoning patterns not present during testing?
2. **Independent Validation**
   - Conceptual soundness; outcome analysis vs. logistic regression baseline
   - **Sensitivity analysis**: response to word-order changes, synonym substitution, missing fields
   - **Disparate impact testing**: adverse action rates by protected class
   - **Adversarial probing**: can a sophisticated borrower manipulate the model?
3. **Governance**

SHAP Interpretability and Adverse Action Notices

SHAP value for feature j in input x (cooperative game theory):

$$\phi_j(x) = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \left[f(S \cup \{j\}) - f(S)\right]$$

Three axioms: **efficiency** ($\sum_j \phi_j = f(x) - \mathbb{E}[f(x)]$), **symmetry**, **dummy**.

Illustrative SHAP waterfall (34-year-old, income $58K, bureau 672, 1 delinquency):

Feature	SHAP ϕ_j	Running PD
Base value	—	$+0.050$
Bureau score $= 672$	$+0.041$	$+0.091$
Delinquencies $= 1$	$+0.027$	$+0.118$
Loan purpose: debt consol.	$+0.019$	$+0.137$
Income $=$ \$58K	-0.012	$+0.125$
Employment: full-time 6yr	-0.015	$+0.110$
Age $= 34$	-0.008	$+0.102$
Final PD		**0.102**

Production Credit Risk System Architecture

Component	Role
Feature Store	Single source of truth; records retrieval timestamp and source per feature; enables point-in-time reconstruction for any historical decision
Inference API	Versioned JSON API: returns raw PD, calibrated PD, reason codes, SHAP values, decision ID
LLM Serving	Self-hosted preferred for credit (full data and model control); API-based creates vendor dependency
Decision DB	Append-only immutable log of every decision; records may be supplemented (outcome) but not overwritten
Monitoring	Data drift (PSI), embedding drift (MMD), score distribution; lagged AUROC by vintage cohort

Human-in-the-Loop and Regulatory Compliance

Three review tiers (required by GDPR Art. 22, ECOA, and SR 11-7):

Tier 1 Clear approvals. Random quality-control sample only.

Tier 2 Borderline cases. Active human review before decision communicated.

Tier 3 Appeals. Full decision record (including SHAP) reviewed; reviewer documents independent assessment.

Automation bias

Research shows reviewers anchor strongly to model recommendations when displayed prominently. Mitigations: show recommendation *after* independent assessment; present SHAP prominently; require documented reasoning before seeing the model's output.

Regulatory documentation package:

- **MDR**: business need, conceptual framework, data, performance metrics, limitations
- **MVR**: independent validation findings and model risk rating

Generating GDPR-Compliant Explanations

GDPR Recital 71 requirement

The controller should implement suitable measures including "the right to obtain human intervention, to express his or her point of view, and to contest the decision." Explanation must be comprehensible to a consumer without financial or technical background.

Plain-language explanation (generated from SHAP waterfall via constrained LLM):

> *"Your application for a €12,000 personal loan was assessed using an automated system. Our system estimated a higher probability of repayment difficulty than our lending criteria allow.*
> **Factors that increased your estimated risk**: *your credit bureau score of 672, which is below the level typically associated with this loan product; a record of one late payment in the past twenty-four months; and a loan purpose of debt consolidation.*
> **Factors that partially reduced your estimated risk**: *six years of continuous full-time employment and an annual income of $58,000.*
> *You have the right to request human review of this decision. To request*

LLM Failure Modes Specific to Credit Applications

1. **Prompt sensitivity**: high sensitivity to word order, synonym substitution, or missing fields is a red flag for instability. Sensitivity analysis is a mandatory SR 11-7 validation step.
2. **Calibration gap**: constrained-decoded probabilities reflect the model's internal representation, not empirical default rates. Calibration is non-optional, not optional.
3. **Protected-class proxies**: LLMs trained on historical financial text may encode correlations between loan purpose language, geographic references, or occupation descriptions and protected classes. Disparate impact testing must include intersectional analyses.
4. **Adversarial manipulation**: a sophisticated borrower may craft application text to trigger favourable model responses. Adversarial probing — testing whether such manipulation is possible — is part of the independent validation.
5. **Vendor lock-in and versioning**: if the provider silently updates the base model, model outputs change without any action by the institution. Versioning contracts and revalidation triggers are required

Summary: The LLM Credit Risk Stack

Layer	Key design choice
Data	Bureau + alternative; FCRA/ECOA/GDPR complian
Preprocessing	LoRA; cost-sensitive loss; text serialisation of missing
Modelling	Sequence classification or constrained generation
Calibration	Platt scaling (small data) or isotonic regression
Household decisions	Persona agents; CRRA utility elicitation
Evaluation	AUROC / KS / Gini; SR 11-7 three pillars
Interpretability	SHAP waterfall → FCRA reason codes
Monitoring	PSI for features; embedding drift for text
Human oversight	Tier 1/2/3 review; automation bias mitigations
Documentation	MDR + MVR + GDPR Art. 22 plain-language

The through-line

Credit AI is not merely a prediction problem — it is a regulated sociotechnical system. Every design choice, from the loss function to the

Appendix

Extended Material: Utility Elicitation, Population Simulation, TTC Validation

Extra / optional material — beyond the core lecture.

Appendix A — Utility-Theoretic Framing

Expected utility: household with wealth W_0 selects:

$$a^* = \arg\max_k \mathbb{E}[U(W_0 + \Delta W_k)]$$

CRRA utility (most common parameterisation):

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}, \quad \gamma > 0$$

$\gamma = 1$: log utility; $\gamma = 2$: moderate risk aversion; $\gamma > 5$: high risk aversion.

LLM-based preference elicitation:

1. Present a sequence of binary lottery choices to identify indifference points
2. Extract the stated indifference points from the conversation
3. Fit CRRA utility to the elicited data by minimising:

$$\hat{\gamma} = \arg\min_\gamma \sum_{j=1}^{J} \left(U_\gamma(c_j) - \lambda_j U_\gamma(g_j) - (1-\lambda_j) U_\gamma(\varepsilon)\right)^2$$

4. Use $\hat{\gamma}$ to evaluate alternatives

Appendix B — Multi-Persona Simulation and Calibration

Synthetic population: run N personas drawn from target population distribution $\mathcal{P}$:

$$\hat{P}(d = k) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[d_i = k]$$

Identifies which subgroups are sensitive to which product features.

Applications:

- **Loan uptake**: estimate take-up rates as function of APR before live A/B test
- **Mortgage suitability**: identify which products create distress risk for which persona types
- **Retirement planning**: test auto-enrolment schemes for low-literacy households

Calibration against survey data (ECB HFCS, US SCF):

1. **Population alignment**: post-stratification weights so joint attribute distribution matches target survey
2. **Behavioural alignment**: compare simulation vs. survey decision

Appendix C — Through-the-Cycle Validation and Backtesting

Two types of temporal model degradation:

- **Concept drift**: the feature $\rightarrow$ default relationship changes as the economy evolves. A model trained on 2015–2019 data may underestimate recession-period default risk.
- **Population drift**: the applicant distribution shifts (new marketing strategy, regulatory change).

Vintage analysis: group loans by origination quarter; track AUROC and KS as outcomes mature; check for time trends in discrimination power.

Point-in-time (PIT) vs. through-the-cycle (TTC):

- Basel III Pillar 1 requires TTC PD estimates (long-run average default rates)
- LLMs trained on recent data tend to produce PIT estimates
- Conversion: apply Platt scaling calibrated on a dataset spanning $\geq$1 full credit cycle (7–10 years)
- The parameters a and b in $\hat{p}_{\text{TTC}} = \sigma(a \cdot f(x) + b)$ absorb both